

Devising a colorimetric aptasensor for detection of basal stem rot-associated RNA marker during early *Ganoderma boninense* infection in oil palm

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ABSTRACT

Basal stem rot (BSR) is a detrimental disease in oil palm (*Elaeis guineensis*) due to the host's lack of symptoms during the early phase of infection. Early detection of BSR in non-symptomatic phase is feasible via molecular-based approaches and has become the major interest among researchers. Our previous study had identified a differentially expressed gene encoding *auxin efflux carrier component 5-like* (*EgPIN5*) at 11 day-post-inoculation of *Ganoderma boninense* (symptomless early stage of infection) on oil palm seedlings. We further evaluated the gene expression on samples collected from an oil palm plantation at different BSR stages (symptomless, mild and severe). Our findings revealed that *EgPIN5* was expressed in symptomless group but was not expressed in either mild or severe BSR groups. Next, we developed a sandwich-type aptamer-based lateral flow assay (LFA) against *EgPIN5* RNA by adapting the versatile DNA-functionalized gold nanoparticles for colorimetric detection. The LFA strip demonstrated visible colorimetric signal of *EgPIN5* RNA at 12.5 ng/μL with limit of detection at 2.45 ng/μL using a scanning device. Based on our consistent results, we presented *EgPIN5* as a diagnostic biomarker for symptomless BSR infection in oil palm. Utilization of LFA assay strip for practical *EgPIN5* RNA detection is hoped to encourage establishment of a fast, cheap, and sensitive point-of-care detection system in plant biology.

1. Introduction

Basal stem rot (BSR) disease is the major causal factor that has drastically reduced palm oil yield, increased agricultural prices, and caused deforestation in Malaysia [1]. The white-rot *G. boninense* is the most pathogenic towards oil palm among the fungal species of *Ganoderma* genus that caused BSR. Hot weather in Malaysia further exacerbated the situation where oil palm yield has decreased by 10–41 % when temperatures rise by 1–4 °C due to water stress [2].

A newly *Ganoderma*-infected palm remains symptomless while the disease progresses slowly. The first sign of BSR appears on foliage when the disease has progressed by 60–70 %, indicating severe phase of infection [3]. Since the host remains asymptomatic, it is critical to implement early detection of *G. boninense* infection. Early detection is

essential so that disease management practices and control measures can be implemented to stop the devastating spread of disease. Different types of BSR detection systems have been employed including real-time polymerase chain reaction (PCR) [4,5], multiplex PCR (mPCR) [6] and enzyme-linked immunosorbent assay (ELISA) [7]. Other methods that are still under evaluation in terms of practicality are multispectral aerial photographs [8], electric nose device [9], hyperspectral imaging and deep learning approach [10]. These detection systems, however, carry drawbacks of high cost, laborious, or low specificity and sensitivity. There is a need to employ a point-of-care detection system that features rapid, cost-effective, high sensitivity and specificity.

Aptamer has gained a mounting amount of attention as biosensors for detection purposes due to the advantages carried by this molecule compared to antibody. It comprises oligonucleotide such as RNA, double- or single-stranded DNA, or protein which can to bind to different

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Abbreviations			
4-PL	4 parameter logistic	AuNPs	gold nanoparticles
<i>EgPIN5</i>	<i>Auxin efflux carrier component 5-like</i>	ITS	internal transcribed region
BSR	Basal stem rot	LFA	lateral flow assay
BLAST	Basic Local Alignment Search Tool	LOD	limit of detection
d.p.i	days-post-incultation	mPCR	multiplex polymerase chain reaction
DEGs	differentially expressed genes	<i>EgPR-1</i>	<i>Pathogenesis-related protein 1</i>
dsDNA	double-stranded DNA	RT-qPCR	real-time quantitative PCR
ELISA	enzyme-linked immunosorbent assay	RWB	rubber wood block
<i>EgEXPB18</i>	<i>expansin B-18-like</i>	ssDNA	single-stranded DNA
<i>EgBGIA</i>	<i>Glu S.griseus protease inhibitor-like</i>	SELEX	Systematic Evolution of Ligands by EXponential Enrichment
		RNA-seq	whole transcriptome

types of molecules. Aptamer is proven to specifically bind to its target with high affinity and specificity. Lateral flow assay (LFA)-based aptasensor is a paper-based platform which utilizes labeled-aptamer as bio-recognition element. It has increasingly become popular for point-of-care analyses in different fields including quality control assessment, biomedicine, food safety, and environmental monitoring due to its low cost of development and feasibility [11,12]. AuNPs are commonly used as label molecules due to the compatibility of this nanomaterials to conjugate with biomaterials like aptamers and its capacity to amplify signal so that the limit of detection (LOD) could be lowered for higher sensitivity of the test strips [13].

Even though PCR-based detection of *Ganoderma* DNA such as that reported by Hilmi et al. (2022) [4] offers high sensitivity, the pathogen DNA may take several months to reach the detection site under field condition. Thus, an earlier indicator of infection such as the plant systemic response marker is preferred. There are a few reports to look for biomarker based on defense response in oil palm against *G. boninense* such as a study conducted by Wulandari and colleague, where a defense-related gene encoding early methionine-labeled polypeptide (*EgEMLP1*) was found significantly upregulated in *G. boninense*-inoculated oil palm ramets from 3 to 7 weeks post infection compared to control [5]. In addition, a study conducted in our laboratory by Nusbah et al. (2016) [14] reported that induced production of metabolites with anti-fungal properties mainly lipids and heterocyclic aromatic organic metabolites in roots of oil palm seedlings were observed during early interaction (from hours to one week of infection) with *G. boninense* suggesting activation of early defense responses in the host plant. It showed that oil palm is responsive to infection, in particular against *G. boninense* from within hours of infection. Our previous study had shown several genes were highly expressed in oil palm during early infection of *G. boninense* via whole transcriptome (RNA-Seq) analysis [15]. Therefore, the differentially expressed genes (DEGs) were further evaluated in this study using field samples to discover potential marker genes during early response against the hemibiotrophic fungal attack. An aptasensor was developed against an RNA fragment of the marker genes via sandwich-type LFA platform, which has paved the way for the development of a fast, cheap, and sensitive point-of-care detection system in plant biology.

2. Material and methods

2.1. Real-time quantitative PCR of RNA-Seq samples

Each sample was a pool RNA from six individual plants under the same group for biological averaging purposes. Primers for the selected genes were designed via Primer3 (v.0.4.0). RT-qPCR reaction mixture consisted of qPCR Green Master Mix LRox (2X) (Biotech rabbit GmbH, Germany), forward and reverse primers, cDNA template and nuclease-free water was prepared as depicted in Supplementary data 1: Table S1. The qPCR reaction mixtures were then operated on Bio-rad

CFX96 Touch Real-Time PCR Detection System (Bio-rad, USA). The thermal cycling parameters of the qPCR were set as in Supplementary data 1: Table S2. Data were analysed by using Bio-Rad CFX Manager (Bio-Rad, USA). Melt curve analysis was applied to determine primer's specificity. The expression levels of all analysed genes were normalized to three most stable reference genes: *GAPDH* 2, β -actin and *NADH* 5. Primers for *NADH* 5 and β -actin that were used have been designed by Ref. [16]. List of all primers used in qPCR were listed in Supplementary data 1: Table S3.

2.2. Field sampling and multiplex PCR

White root samples were harvested from oil palm plantations with different BSR conditions. Samples of group A, B, and C were harvested from a plantation in Keratong Pahang, Malaysia whereby group A was appointed to healthy-looking, no foliar symptom and fruiting body, and still producing fruits. Whereas group B was considered mild BSR symptoms with fruiting bodies detected at the base, having foliar symptom but absence of observable stem rotting, and still producing yield. Group C was considered severe BSR symptoms with fruiting bodies at the base, severe foliar symptoms of BSR (unopened first and second leaves and mid fronds collapse at the petiole) and stem rotting observed and not producing yield. Group D was harvested from Hulu Paka Terengganu where there was no previous report of BSR incidence at the region [17]. Harvesting of samples and RNA extraction was performed following the methods described in Ref. [15]. Reverse transcriptase mPCR was conducted using three sets of primers: *EgEXPB18*, *EgPIN5*, and *EgGAPDH* (Supplementary data 1: Table S4). Reaction setup and cycling protocol are available in Supplementary data 1: Tables S5 and S6 respectively.

2.3. SELEX strategy for aptamer selection

Each round of SELEX consisted of several consecutive methods including hybridization, emPCR, recovery of PCR product, and lambda exonuclease digestion. Ten rounds of SELEX were performed to obtained high affinity and specificity aptamer candidates towards the target molecules.

2.3.1. Single-stranded DNA random sequences library

DNA random library was synthesized (Integrated DNA Technologies, Iowa, USA) as 77-mer single stranded oligonucleotides where the sequences are: GCCTGTTGTGAGCCTCTGTCGAA-N₃₀-TTGAGCGTT-TATTCTGTCTCCC. The central 30 bp sequence consisted of random nucleotide (N₃₀), where every base position having equimolar concentrations between A, T, G, and C. The sequences that flanking the random sequence were primer binding sites for amplification purposes. Primer set used to bind at the primer sites were as the following: 5'-GCCTGTTGTGAGCCTCTGTCGAA-3' (forward) and 5' phosphorylated-GGGAGACAAGAATAAACGCTCAA-3' (reverse). The 5'-end of the

reverse primer was modified with phosphorylation for subsequent mononucleotide removal from double-stranded DNA (dsDNA) to yield single-stranded DNA (ssDNA) by lambda exonuclease.

2.3.2. Counterselection library setup

Counterselection library was prepared according to Ref. [18] with modifications. Briefly, 1 nmol of initial library was heated at 98 °C for 1 min, then cooled to 4 °C for 1 min. Thirty microliter of the library was mixed with 20 µL R buffer (20 mM HEPES, 20 mM sodium acetate, 140 mM potassium acetate, and 3 mM magnesium acetate, pH 7.4) and incubated for 5 min at room temperature. Another tube containing 50 µL of 100 µg streptavidin-coated magnetic beads in R buffer was incubated for 10 min at room temperature. Next, both tubes were mixed and incubated for 15 min at room temperature. After incubation, magnet was applied to pellet the magnetic beads carrying their bound aptamers. Supernatant was collected into a new tube to proceed with hybridisation with target. The magnetic beads-bound aptamer was eluted with 80 µL nuclease-free water via heating at 80 °C for 45 s. The amount of these aptamer was measured by NanoDrop™ Lite Spectrophotometer (Thermo Scientific, USA) for monitoring its evolution across SELEX rounds.

2.3.3. Hybridisation of random library with RNA target

The counterselection library was heated to 98 °C for 1 min, while RNA target was heated at 65 °C for 3 min. Both tubes were cooled down to 4 °C for 1 min and incubated at RT for 5 min. Five hundred pmol of library was mixed with 50 pmol of RNA at RT in 100 µL of R buffer for 10 min. Streptavidin-coated magnetic beads (150 µg) was added and incubated for 10 min at RT. Magnet was applied to pellet the magnetic beads carrying the RNA-bound aptamers. Supernatant was discarded and the pellet was washed thrice with R buffer. Aptamer was eluted with 50 µL of water through applying heat at 75 °C for 40 s.

2.3.4. Emulsion PCR

Oil emPCR was performed according to method described by Ref. [19] with some modifications. The oil phase was formulated by using mineral oil containing 4.5 % Span 80, 0.4 % Tween 80, and 0.05 % Triton X-100 (Sigma Aldrich, USA) and vortexed vigorously for at least 5 min. Whereas the aqueous phase (PCR mixture) was prepared using *Taq* DNA polymerase (New England Biolabs, USA) as tabulated in Supplementary data 1: Table S7. Water-in-oil emulsion was prepared by adding dropwise of 200 µL of ice-cooled PCR mixture into continuously stirred (~1400 rpm) 400 µL oil phase in a HPLC vial for 2 min by using a micro stir bar on a plate stirrer (Fisher Scientific, USA) and continued stirring for another 5 min. The emulsion was subsequently aliquoted into PCR tubes (50 µL/tube). Total volume of emulsion needed to generate sufficient amount of ssDNA library for next round of SELEX was pre-determined. The emulsion was then subjected to PCR cycling protocol as described in Supplementary data 1: Table S8).

2.3.5. Recovery of PCR product

Recovery of PCR product from emulsion was carried out according to Ref. [20] with minor modifications. Following the PCR reaction, emulsion was aliquoted into 1.5 mL tube (300 µL per tube). The tubes were then centrifuged at 16,000×g for 15 min. The top oil phase layer was carefully removed, leaving an aqueous phase containing the PCR product at the bottom. Oil residue was washed with water-saturated diethyl ether (DEE). The tube was briefly vortexed and top DEE layer was discarded. Remaining DEE residue was removed via vacuum-concentrating in Eppendorf® Concentrator 5301 (Eppendorf, Germany) for 15 min. PCR product was purified from the aqueous phase by using Monarch® PCR & DNA Cleanup Kit (5 µg) (New England Biolabs, USA) according to manufacturer's protocol.

2.3.6. Lambda exonuclease digestion to yield single-stranded DNA aptamer library

The double-stranded PCR amplicon undergone lambda exonuclease (Thermo Scientific™, USA) digestion to prepare ssDNA random sequence library for next round of SELEX. Briefly, 50 µL of total reaction containing 1X lambda exonuclease reaction buffer, 10 U of lambda exonuclease enzyme, PCR product (DNA up to 2 µg), and nuclease-free water. The reaction mixture was incubated at 37 °C for 30 min, followed by heat inactivation at 80 °C for 10 min. Upon completion of the enzymatic digestion, the aptamer library (ssDNA) was subjected to oligonucleotide clean-up through Monarch® PCR & DNA Cleanup Kit (New England Biolabs, USA) according to manufacturer's protocol.

2.3.7. Cloning of aptamer candidates

Sequences of aptamer candidates were discovered via Gateway Recombination Cloning Technology (Invitrogen, USA) by following the manufacturer's protocol. Vector pDONR™221 carrying aptamers sequences from BP recombination reaction was introduced into One Shot® OmniMAX™ 2 T1R Chemically Competent Cells. Upon overnight incubation of transformed *E. coli* on LB plates, 30 individual colonies were randomly picked and proceeded with plasmid purification using QIAprep® Spin Miniprep Kit (Qiagen, Germany) according to manufacturer's protocol.

2.3.8. Sanger sequencing and sequence alignment

Sequencing of the cloned aptamers was completed by service provider via Sanger sequencing [21]. The sequencing was performed from one direction, which was from 5'-end towards 3'-end using standard M13-F forward primer. The sequences of aptamer candidates were aligned using MAFFT alignment algorithm [22,23] with default parameters in Geneious R9 package (<http://www.geneious.com>) [24]. Consensus sequences was identified from the alignment.

2.4. Development of LFA strip

2.4.1. Preparation of AuNPs

AuNPs was prepared by heating 250 ml of 0.01 % w/v HAuCl₄ until boiling and vigorously stirred in 500 ml flask. Next, 4.5 ml of 1 % (w/v) trisodium citrate was added immediately into the boiling solution. The solution turned deep blue within 20 s, before changing to wine-red after 60 s. Boiling was continued for 10 min. The AuNPs were cooled to room temperature with continuous stirring for another 15 min. The colloidal AuNPs were kept in a dark bottle at 4 °C. The colloidal AuNPs were imaged using transmission electron microscope (TEM) following in-house method to estimate its size and shape. The concentration of synthesized AuNPs was estimated following the Beer-Lambert law: $C = A / (L \times \epsilon)$, where A corresponds to absorbance at 520 nm, while L refers to optical length, and ϵ refers to molar extinction coefficient.

2.4.2. Synthesis of aptamers

Both DNA and RNA used in the study were synthesized by Integrated DNA Technologies (IDT), USA. Selection of aptamer candidates was made randomly from sequenced aptamers, given that every cloned aptamer has affinity towards RNA target as they secured to the final round of SELEX. Three aptamer candidates were synthesized for primary and secondary aptamer. Apseq14, Apseq26, and Apseq50 were assigned as primary aptamers (hence called PriApseq14, PriApseq26, and PriApseq50), whereas Apseq3, Apseq16, and Apseq20 were assigned as secondary aptamers (hence called SecApseq3, SecApseq16, and SecApseq20). Primary aptamers were synthesized with thiol linker at 5'-end. Secondary aptamer and control oligonucleotides (Cont14, Cont26, and Cont50) were linked with biotin at 3'-end. The biotin linker will be used for binding with streptavidin to aid in immobilization at the test and control zone of the LFA strip. The control oligonucleotides were base complementary to the primary aptamer accordingly. RNA as target was synthesized without any linker. Their sequences and terminal

modifications were presented in Table 1.

2.4.3. Preparation of primary aptamer-AuNPs conjugates

Conjugation of thiol linked-primary aptamer with AuNPs was performed via freeze-directed construction as described by Ref. [25]. Four microliter of 100 μM aptamer was mixed with 100 μL of 10 nM AuNPs. The mixture was briefly resuspended with micropipette and subsequently incubated in −20 °C for 2 h. After incubation, the conjugate was thawed at room temperature prior to centrifugation at 13,000 rpm for 13 min at RT. Five hundred microliters of HEPES buffer (5 mM, pH 7.6) was used to wash the primary aptamer-AuNPs conjugate thrice before resuspended in the same buffer.

2.4.4. Preparation of secondary aptamer- and control oligonucleotide-streptavidin conjugates

The biotinylated secondary aptamer and control oligonucleotide were conjugated with streptavidin following [26] with slight modification. Briefly, the aptamer or control oligonucleotide in 3X SSC buffer was mixed with streptavidin in 1X PBS buffer with 1:1 M ratio, briefly vortexed and incubated for 2 h at RT to undergo conjugation process.

2.4.5. Fabrication of lateral flow strip

LFA strip for detection of EgPIN5 RNA target molecules were formed from nitrocellulose membrane (Hi-Flow Plus HF120 Merck, Germany) that will be used to construct the test and control zones, and cellulose fibre (SureWick C083 Merck, Germany) as sample application and absorbent pad. The length of nitrocellulose membrane, sample pad and absorbent pad were 20 mm, 18 mm and 22 mm respectively. Sample application pad was initially prepared by soaking in a buffer containing 0.1 M Tris pH 8.0, 3 % sucrose, 1 % BSA, and 0.05 % Tween 20 for 2 h before subjected to drying overnight at RT. The secondary aptamer- and control oligonucleotide-streptavidin conjugates (0.2 μL) were applied on the nitrocellulose membrane to form test and control zones, respectively in a dotlike shape by using micropipette. Distance between the zones was 4 mm. The pad was then incubated overnight at 40 °C. The prepared sample application pad, nitrocellulose membrane, and absorbent pad were subsequently assembled on a laminated backing card. The overlapped regions between pads were 2 mm. Finally, the LFA construct was cut into 4 mm width strips (Fig. 1).

2.4.6. Lateral flow assay

Before the LFA test begins, 4 μL of primary aptamer-AuNPs conjugate were applied onto sample application pad. RNA target with desired concentration in 120 μL of 15X SSC buffer (running buffer) was prepared in 96-well microplate. Next, the strip was dipped into the wells and kept slanted. The capillary actions of the strip create a flow of the target in running buffer through the nitrocellulose membrane and absorbent pad. After 15 min, the signal of test and control zone were recorded with Epson Perfection V370 Photo Scanner (Epson, Japan) at 600 dots per inch (dpi) resolution. The signal was analysed quantitatively by using ImageJ image processing tools [27]. LOD was determined using RNA concentration from 200 ng/μL to 1.56. The data was plotted into an XY

graph style using Graphpad Prism (San Diego, USA). Every concentration was performed in triplicates.

2.5. LFA using field RNA samples spiked with synthetic RNA fragment

One hundred and 20 μL of RNA extracts from Keratong and Hulu Paka plantation were assayed with the LFA strips at concentration of 100 ng/μL. Next, 120 μL of 100 ng/μL of samples A11 was spiked with 6.25 and 200 ng/μL of synthetic RNA fragment and applied on the LFA using similar technique as previously explained. The signal was measured by using ImageJ image processing tools.

3. Results

3.1. Quantitative PCR of EgPR1, EgEXPB18, EgBGIA, and EgPIN5

Our previous study has performed high-throughput transcriptome analysis via RNA-Seq and discovered DEGs in oil palm during early phase of *G. boninense* infection. Several genes from different functional groups were identified to be significantly regulated in *G. boninense*-treated samples compared to untreated control from three to eleven day-post-inoculation (d.p.i). Analysis from the RNA-Seq data prompted us to select genes which are having potential as biomarkers during early *G. boninense* attack in oil palm. The first factor that influenced our selection is genes where their expression was significantly altered in *G. boninense*-treated samples versus untreated control based on RNA-Seq data. *G. boninense*-treated sample refers to samples that had been treated with *G. boninense*-colonized rubber wood block (RWB). Pathogenesis-related protein 1 (*EgPR1*), expansin B-18-like (*EgEXPB18*), *Glu S.griseus* protease inhibitor-like (*EgBGIA*), and *EgPIN5* were selected as the candidate marker genes since they were among the top upregulated genes (RNA-Seq data). Earlier, the *EgPIN5* was annotated as *EgPIN8* in the database, and so was reported as such in our previous work, but then revised and annotated as *EgPIN5* in the database. Second factor taken into consideration is these genes must show significant difference in mock samples (treated with bare RWB) compared to *G. boninense*-treated samples. While, the candidate genes must also show insignificant different expression between mock-treated samples and untreated control. In this study, expression of the candidate genes was determined on control, mock and *G. boninense*-treated samples via reverse transcription quantitative real-time PCR (RT-qPCR). Each sample consisted of pool of six plants (untreated, mock, or treated samples) as a biological averaging approach.

The RT-qPCR was done using similar samples that were used for RNA-Seq in our previous study [15]. As shown in Fig. 2, all genes exhibited higher expression in *G. boninense*-treated samples compared to mock samples. Expression of *EgPR1*, *EgBGIA*, and *EgEXPB18* in *G. boninense*-treated samples were highest at 3 or 7 d.p.i, and subsequently reduced to the lowest and insignificant expression at 11 d.p.i. We suggested that the trend of expression of these three genes may be further reduced after 11 d.p.i. On the other hand, *EgPIN5* showed increasing pattern of expression until reached the highest at 11 d.p.i. We

Table 1
Sequences of aptamer candidates and EgPIN5 RNA fragment.

Label	5' Modification	Sequence	3' Modification
PriApseq14	/5ThioMC6-D/	CGACAAAACCATCATGATCACTTGTAACGG	
Cont14	/5Biosg/	CCGTTACAAGTGATCATGAT	
PriApseq50	/5ThioMC6-D/	ATTACCATCATGATCACTTGGGTGACTCGG	
Cont50	/5Biosg/	CCGAGTCACCCAAGTGATCA	
PriApseq26	/5ThioMC6-D/	CGTATGCATCCCCATCATGATCACTAGCCG	
Cont26	/5Biosg/	CGGCTAGTGATCATGATGGG	
SecApseq20		CATGTACCATCATGATCACTCCGTCCTCG	/3Bio/
SecApseq3		CGCCATCCCACTAGAGTTGTTGCCACCACC	/3Bio/
SecApseq16		CAAATACCATCATGATCACTTACCCGCTG	/3Bio/
EgPIN5 RNA		UUAAGGAAGGCAAGAAUACAGGAAGCUAUGGUGGUGGCCAUCUCUAGUGAUGAUGGU	

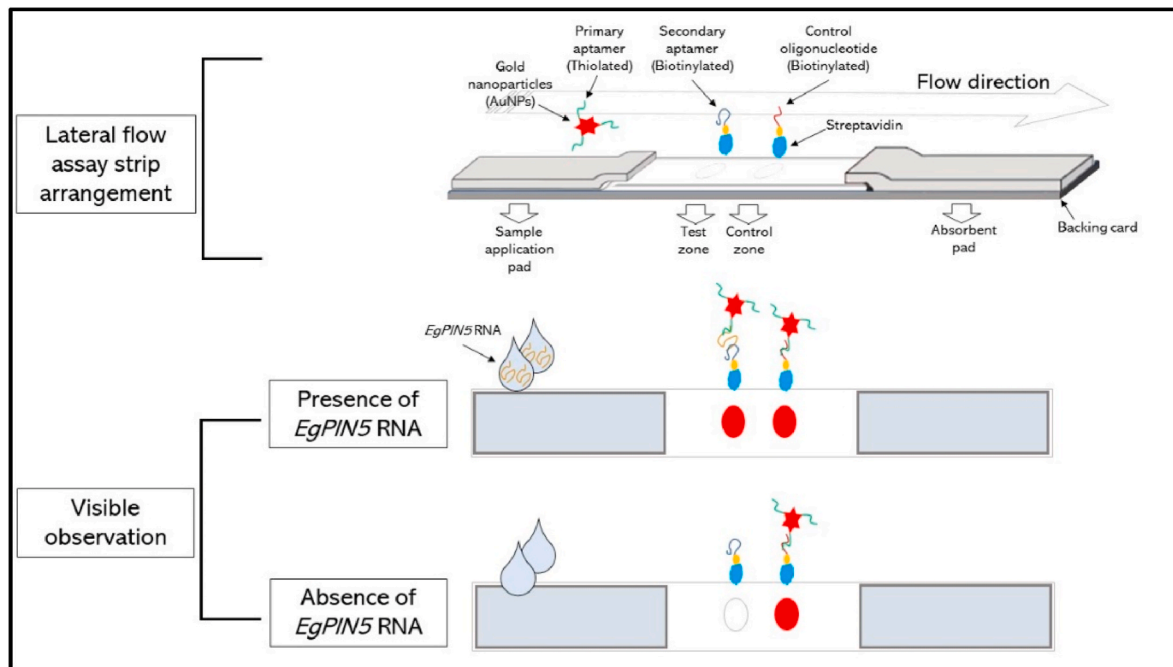


Fig. 1. Schematic diagram of lateral flow assay strip targeting EgPIN5 RNA.

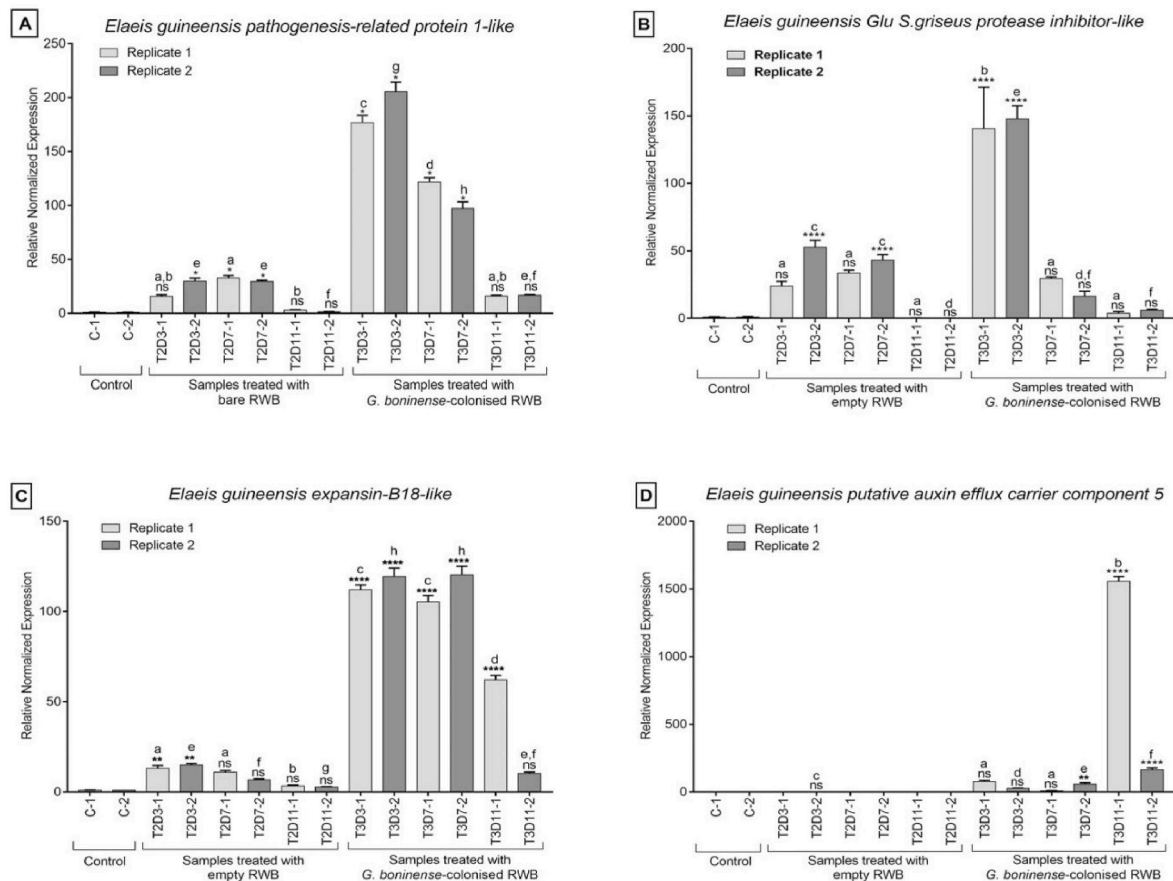


Fig. 2. Expression of candidate genes in *Ganoderma boninense*-treated samples compared to mock samples (empty rubber wood block). The expressions of each gene were normalized by reference genes; *glyceraldehyde 3-phosphate dehydrogenase 2* (GAPDH 2), *nicotinamide adenine dinucleotide + H 5* (NADH 5) and β -actin expression levels. Data are expressed as the mean $\pm$ SEM of three individual technical replicates of each sample. * $P < 0.01$ is significantly differed compared to corresponding control as assessed by one-way ANOVA analysis followed by Tukey's test. ns is not significant. Different superscript letters between samples (within replicate group) indicate significant different ($P < 0.01$) in mean values. RWB: Rubber wood block.

do not have any data regarding expression of *EgPIN5* beyond 11 d.p.i., however *EgPIN5* could be a potent biomarker if the expression keeps increasing. We have proven that the artificially infected oil palm seedling seedlings that expressed *EgPIN5* at the very early stages of infection developed the symptoms of BSR after the 24th weeks of infection, as reported in our previous work [15]. The diseased plants showed wilted leaves and basidiocarps at its bole. Even though the *EgPIN5* expression data was not obtain from the same palm (due to destructive sampling), but the diseased palms were among the biological replicates of the harvested samples. In addition, *EgPIN5* surprisingly did not show expression in mock samples, thus proposing that the genes discriminatingly responded to pathogenic (*G. boninense*) stress only. The mock samples of *EgPR1* and *EgBGLA* demonstrated significant expression of more than thirty-fold compared to control, but *EgEXPB18* was expressed at relatively lower level and insignificant at 7 and 11 d.p.i. *EgEXPB18* and *EgPIN5* were subsequently selected for evaluation on field samples.

3.2. Multiplex PCR of candidate marker genes on field samples

mPCR enables multiple target genes being simultaneously detected in a single reaction using separate pairs of primers. This method enables high-throughput, which saves time and cost compared to endpoint PCR.

Expression of *EgEXPB18* and *EgPIN5* was assessed via mPCR on RNA obtained from root of oil palm trees in oil palm plantations with different incidences of BSR. An internal control, *EgGAPDH* was included in the assay for validity purposes. The primer pair for each gene was carefully designed to be specific against the gene-of-interest only, and avoiding the tendency of forming dimers. Besides, a distinctive length of the amplified product is ensured as they electrophoresed on agarose gel. The shortest amplicon length was *EgGAPDH* (218 bp), followed by *EgPIN5* (409 bp), and *EgEXPB18* at 650 bp. The mPCR was performed with 20 μ L total volume containing 200 nM of each primer, 10 ng of cDNA template and PCR cycling protocol setup for 28 cycles.

RNA from samples of group A, B, and C were harvested from an oil palm plantation in Keratong Pahang, Malaysia where healthy-looking, mild and severe BSR symptoms were observed, respectively. Whereas group D was collected from BSR-free plantation in Hulu Paka Terengganu, Malaysia. Findings revealed that one sample from group A (A11) showed *EgPIN5* amplicon, hence denoted expression of the gene (Fig. 3), but the gene was not detected in any samples of group B, C, and D. In contrast, *EgEXPB18* amplicons were observed in groups (A, B, C, and D) samples with variable extent of expressions (band intensities) with the exception of sample D18 which was believed to be degraded (Fig. 3B). The internal control, *EgGAPDH* was expressed on all samples of

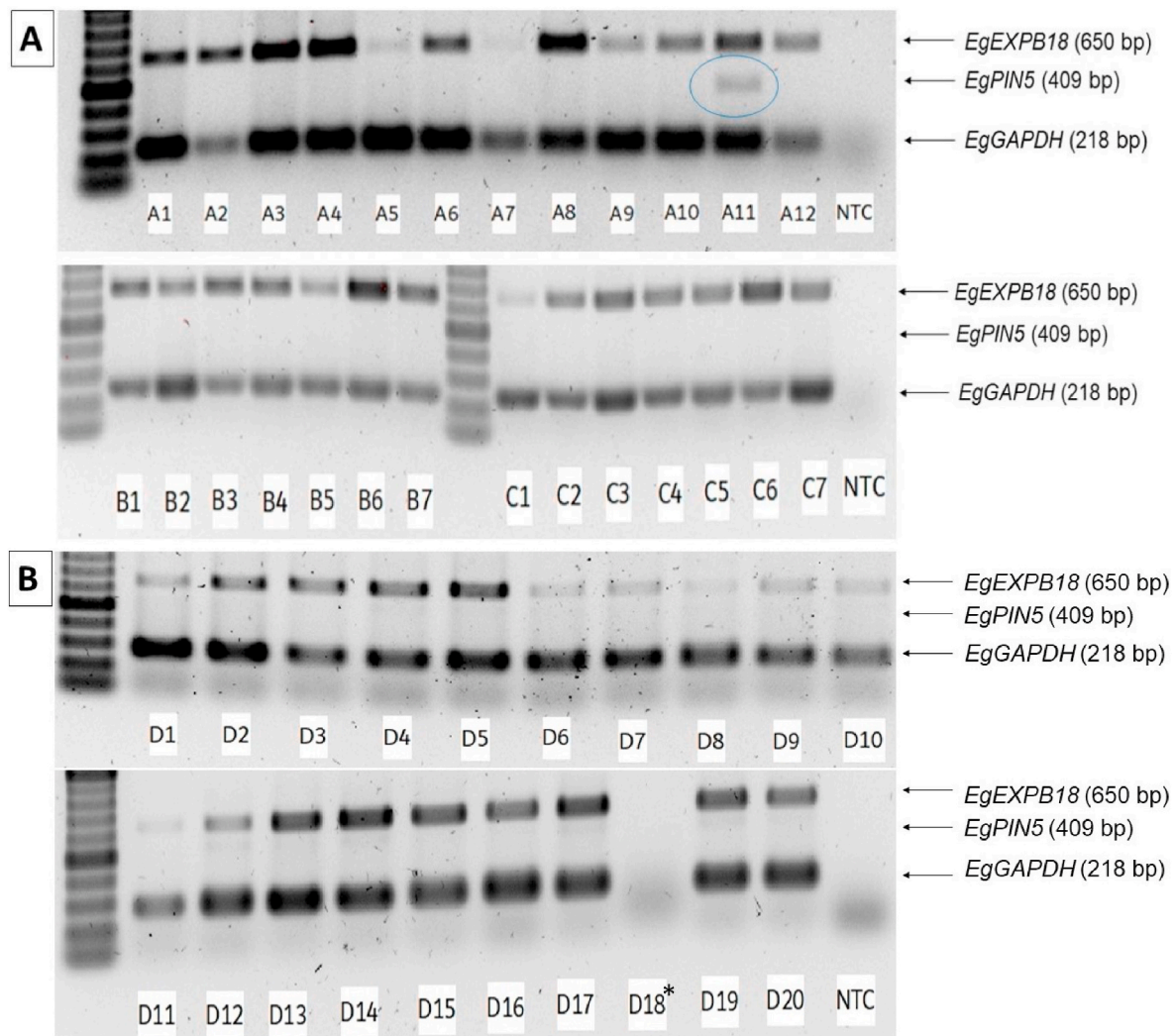


Fig. 3. Multiplex PCR of glyceraldehyde 3-phosphate dehydrogenase 2 (*EgGAPDH*), expansin B-18-like (*EgEXPB18*), and auxin efflux carrier component 5-like (*EgPIN5*) performed on oil palm field samples. Total RNA was isolated from (A) samples of group A, B, and C which were obtained from oil palm plantation at Keratong Pahang, Malaysia and (B) samples of group D were from oil palm plantation at Hulu Paka Terengganu, Malaysia. Blue circle denotes *EgPIN5* amplicon on samples A11. *Samples D18 was considered degraded.

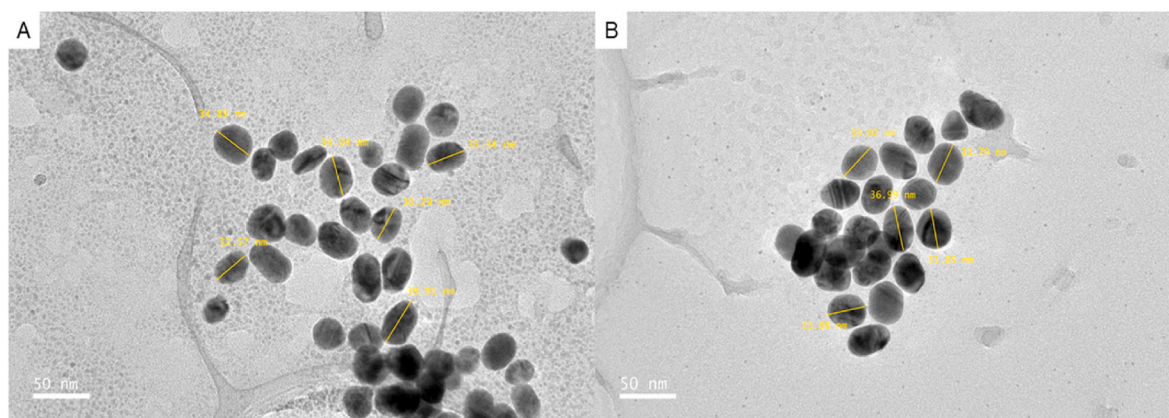


Fig. 5. Colloidal gold nanoparticles (AuNPs) under transmission electron microscope. (A) Bare AuNPs. (B) Primary aptamer-coated AuNPs. The AuNPs were prepared using citrate reduction of HAuCl₄ under reflux. Conjugation of thiol linked-primary aptamer with AuNPs was performed via freeze-directed construction. Average size of bare and primary aptamer-coated AuNPs were determined as 34.1 nm and 34.5 nm respectively.

captured by the primary aptamer at sample application pad. Capillary action of the cellulose fibre and nitrocellulose membrane drew the running buffer carrying the target-bound and free (unbound) primary aptamers through nitrocellulose membrane until reach the end part of the strip (absorbent pad). At test zone, the primary aptamer-bound RNA will be immobilized through captured by secondary aptamer hence resulted in colour signal built up.

Free primary aptamer and uncaptured analytes continue their flow to the next part. At control zone, the free primary aptamer will be captured by control oligonucleotides via complementary base pairing and resulted in colour build up. The colorimetric 'dotlike' signal at the test and control zone can be observed as early as 1 min by naked eye. The colour intensity of the test and control zones were recorded after 15 min of assay using a normal scanner and data was analysed by ImageJ image processing tools.

Nine possible combinations between primary and secondary aptamers were evaluated in the sandwich-type LFA to find which combination elicited the highest signal. Combination between 14/20 (PriApsq14 and SecApsq20) was recorded elicited the highest signal (76.02 px²), followed by 14/3 (53.76 px²), 26/3 (47.98 px²), 14/16 (29.82 px²), 26/20 (18.79 px²), and 14/16 (9.60 px²) (Fig. 6). Combinations using PriApsq50 with all secondary aptamers yielded extremely low signal which were less than 2 px². Control zone for every combination showed constant signal intensity (68 px² mean with 2.5 px² standard deviation) which validated that the LFA strip is working.

3.5. Determination of LOD and field sample tests

PriApsq14 and SecApsq20 combination was further proceeded for LOD using a range of target concentrations. *EgPIN5* synthetic RNA was prepared from 200 ng/μL with two-fold serial dilutions until 1.56 ng/μL. The recorded signals followed dose-dependent manner, where increase concentration of RNA target resulted in increased signal intensity. The data of recorded signals were plotted into an XY graph mode using Graphpad Prism software (San Diego, USA). Signals elicited at low to high concentration of target showed a sigmoidal profile. Hence, regression curve was generated following sigmoidal 4 parameter logistic (4-PL) equation [30]. Correlation coefficient obtained from the 4-PL curve was 0.9648 (Fig. 7). The LOD of the assay was calculated at 2.45 ng/μL. This value was obtained by using three times the value of standard deviation of signals at the lowest concentration (1.56 ng/μL), then the value was being interpolated into the 4-PL equation [30]. The range of target concentrations where the red dot signal could be viewed with naked eye were from 12.5 ng/μL to 200 ng/μL.

Next, RNA extracts from field samples were applied on the LFA strips to test whether real samples from field could generate signal on the strips. One hundred 20 μL of 100 ng/μL of the extracts failed to generate any signal on the strips from all groups of BSR stages (Supplementary data 1: Fig. S3). We further spiked sample A11 (positive samples for *EgPIN5* detection by mPCR) with 6.25 ng/μL and 200 ng/μL of synthetic *EgPIN5* RNA fragment to observe the colorimetric signals and monitor presence of inhibitors in the extract. Interestingly, results shown that non-spiked A11 did not show any signal, but both spiked extracts

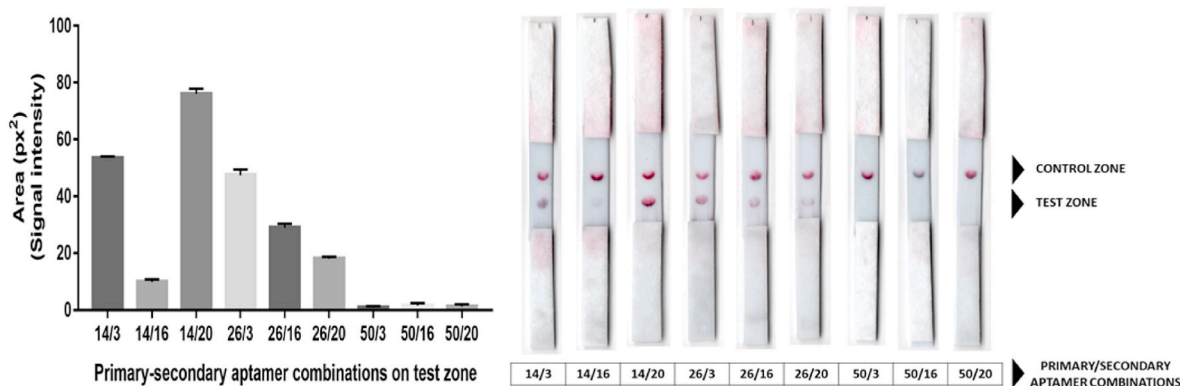
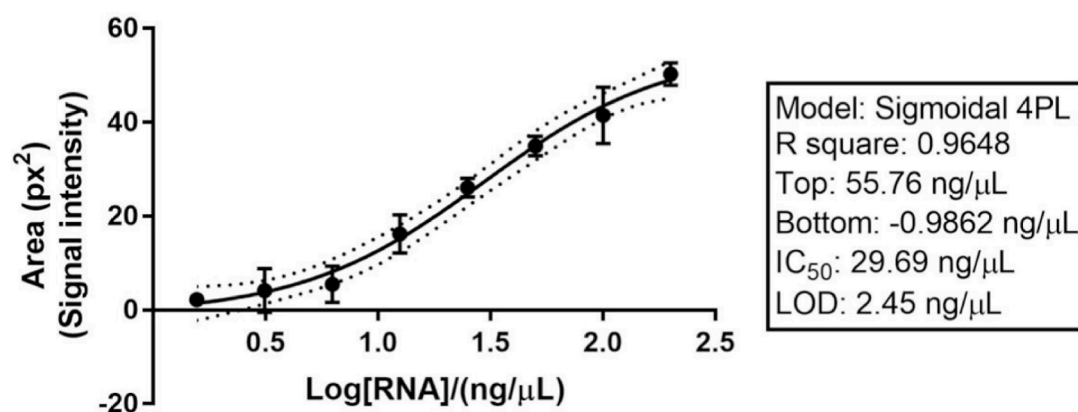
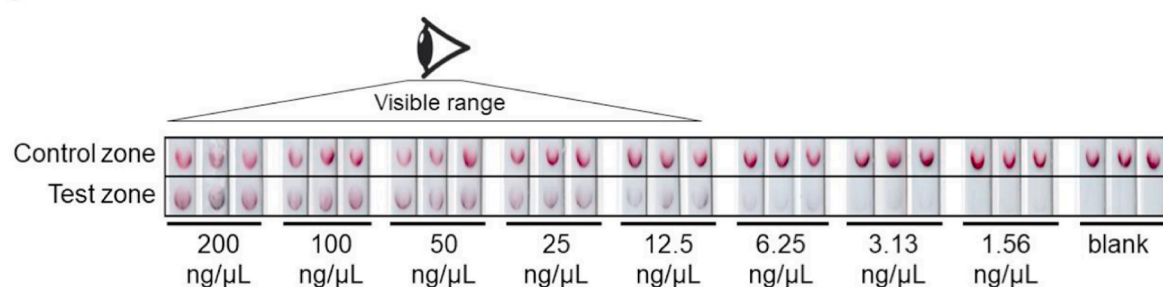


Fig. 6. Optimization of primary and secondary aptamer combinations in aptamer-based lateral flow assay strip for detection of *Auxin efflux carrier component 5* (*EgPIN5*) RNA fragment. Each combination is assayed with 200 ng/μL of *EgPIN5* RNA fragment in three replicates. The histogram on the left side representing the intensity of the dotlike colorimetric signal appeared on the strips (right) as recorded and processed with ImageJ software.

A



B



C

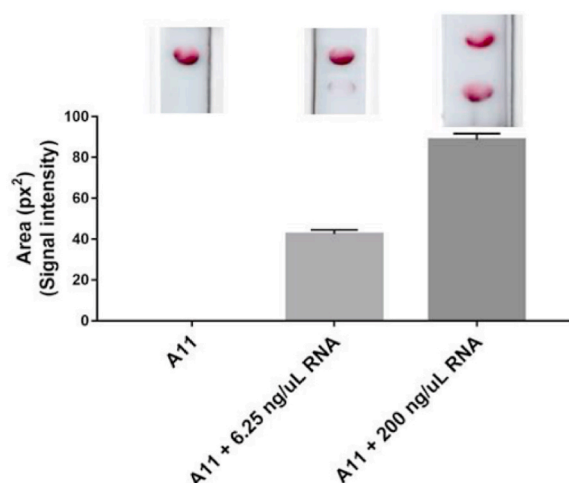


Fig. 7. Limit of detection (LOD) of RNA target and field samples spiked with RNA target. The LFA strips were tested with 120 μ L of 1.56–200 ng/ μ L of *EgPIN5* RNA fragment. Regression curve was generated following sigmoidal 4 parameter logistic (4-PL) equation (**A**). The signal was visible from 12.5 ng/ μ L to 200 ng/ μ L (**B**), while LOD was determined at 2.45 ng/ μ L. Next, the pure RNA extract from field samples (sample A11) were tested with the LFA strips at 100 ng/ μ L. Similar concentration of the RNA extracts were also spiked with 6.25 ng/ μ L and 200 ng/ μ L of synthetic *EgPIN5* RNA and tested on the LFA strips. The spiked extracts produced signals in parallel with signals generated by RNA synthetic alone (**C**). All assays were performed in triplicates.

generated signals that corresponding to signals assayed by similar concentration of synthetic RNA alone (Fig. 7C).

4. Discussion

Detecting BSR biomarkers at the early phase of *G. boninense* infection in the field based on expressed genes from host defense response is preferred rather than pathogen DNA. Plants employ sophisticated

signalling pathway to interpret the newly invading pathogenic fungi as threats and spread the signal to whole plant parts in a short period of time (within a few days) through systemic response [31] once there is plant pathogen interaction. Detection of BSR using *Ganoderma* spp. genomic DNA markers such as the ribosomal internal transcribed region (ITS) conserved region for early phase infection takes much longer for obtaining a positive result following pathogenic fungal infection possibly months as the fungal mycelia must reach the detection site.

However, the process of selecting oil palm biomarkers for specific response towards BSR infection is challenging. A high-throughput RNA-Seq approach which enables whole transcriptome analysis is required. Several DEGs from RNA-Seq data of oil palm root at early infection phase of *G. boninense* from our previous study were interpreted as responding against the *G. boninense* early infection [15].

EgPR1 is one of the major targets as biomarker in this study because of its high transcript abundance, positioned among the top most up-regulated genes [15]. Additionally, *EgPR1* plays pivotal function in the host plant where the gene encodes an antimicrobial protein that can bind and sequesters sterol from pathogen which subsequently inhibit pathogen's growth. In fact, this gene was chosen as a marker for plant immune signalling [32]. However, appointing *EgPR1* as specific biomarker is ambitious because the sequence is homologous with other transcript variants. Basic Local Alignment Search Tool (BLAST) applied using the *EgPR1* sequence (from our RNA-Seq data) as query and returned 0.0 E value, 100 % coverage and more than 93 % similarity with other nine *EgPR1* transcript variants (Supplementary data 1: Table S10). Moreover, mock-treated samples showed significant expression of the gene compared to control, hence causing the development of robust and specific colorimetric LFA detection kit troublesome.

Glu S.griseus protease inhibitor was previously reported to be involved in potato defense responses against herbivory attack [33]. When it binds to serine proteinases like trypsin and chymotrypsin in the gut, digestion is interrupted leading to further interruption of developmental process therefore increasing mortality rate of the herbivores. In our study, the gene was significantly expressed in *G. boninense*-treated sample at 3 d.p.i but reduced significantly at 7 and 11 d.p.i. Furthermore, mock-treated samples showed relatively high expression of *EgB-GIA* compared with other candidate genes (*EgPR1*, *EgEXPB18*, and *EgPIN5*). Hence, the process of appointing the gene as biomarker was halted because the gene was considered responsive to abiotic stresses.

Expansin is involved in cell wall expansion and loosening [34]. This gene was studied for its function in plant stress response against abiotic stresses [35,36] and necrotrophic attack of *Botrytis cinerea* on *Arabidopsis thaliana* [37]. Expansin was known to enhance host susceptibility against pathogen through promoting breakage of non-covalent bonds that hold the structure of polysaccharides exposing larger surface glucans for cellulase attack [38]. In our study, the *EgEXPB18* was highly expressed in *G. boninense*-treated samples compared to untreated control as shown by RNA-Seq and validated by qPCR. Moreover, its expression in mock-treated samples was significantly different from *G. boninense*-treated samples at all time points. Nevertheless, assessment performed on oil palm plantation samples demonstrated that this gene was shown to be expressed in samples with all BSR conditions including healthy, mild and severe at the Keratong plantation as well as in BSR-free samples at the Hulu Paka plantation, implying that the gene may confer non-specific responses towards stresses.

Auxin efflux carrier component regulate polar auxin transport in root cells. With the aid of protein kinase PINOID (PID) polar trafficking of auxin out of root cells inhibit the auxin signalling and root hair growth [39]. Upregulated expression of *EgPIN5* may cause intensive efflux of auxin from cells in oil palm root, potentially enhancing the susceptibility of the host. In this study, the absence of *EgPIN5* amplification in mock-treated samples offered high potential of the gene to be developed as biomarker for early infection of *G. boninense* in oil palm.

Assessment of the LFA strips with RNA extracts from field failed to generate colorimetric signal even at 100 ng/ μ L of the extracts. We proposed that it was due to low load of the RNA copies in the field samples. However, spiked A11 at low and high concentrations of synthetic RNA resulted in signals parallel to signals generated by synthetic RNA alone thus implying that application of field samples did not interfere with development of colorimetric signal on the LFA strip. The aptamer developed against the *EgPIN5* RNA and optimization of LFA prototype in this study was not affected by any interfering agents which

could be present in the field extracts such as inhibitors or the RNA pool itself.

We further recommended application of isothermal amplification technique which is feasible to be carried out in the field such as transcription-mediated amplification (TMA) [40] in the future to amplify the low load of the target RNA copies required to meet the amount for LOD and producing visible signals on the LFA strips. TMA is a method to amplify cellular DNA, mRNA, viral RNA, or highly structured ribosomal RNA. This method has the power to amplify less than 10 copies of mRNA or less than 50 copies for rRNA template by using reverse transcriptase and RNA polymerase. TMA has been widely applied in clinical applications including diagnostics of SARS-CoV2 [41], and sexually-transmitted microbes [42,43]. Furthermore, TMA does not require the thermal cycling protocol like PCR because it can be performed in isothermal mode at 37–42 °C. Hence, this method is considerably suitable to be performed on-site without using high-end instrument for amplification of *EgPIN5* RNA.

Besides, another method which could be employed for low load of RNA target is Sensitive Splint-based One-pot Isothermal RNA detection, termed SENSR. It is a specific technique used for the detection of RNA at low concentrations. SENSR is designed to be highly sensitive and can be used to detect RNA without the need for complex and expensive laboratory equipment. It involves a one-pot isothermal amplification process to amplify RNA and a splint-based design to enable the detection. SENSR is a promising technology for applications such as molecular diagnostics and pathogen detection. For example, Woo et al. (2020) [44] developed a highly sensitive and specific one-pot assay for fluorescence-based detection of RNA from SARS-CoV-2. Utilizing ligation-dependent method (SENSR) in combination with isothermal amplification to establish a one-pot RNA detection method, the group managed to reach a limit of detection of 0.1-aM (10^{-18} M) of RNA concentration.

Based on consensus sequence 'CCAT CATG ATC' that were present in 88 % of the clones, the aptamers were proposed to bind on the RNA target via Watson–Crick complementary base-pairing because observation on the RNA target sequence revealed reverse-complement bases corresponding to the consensus sequence at position 49 to 59 as depicted in Fig. 8. However, none of our cloned aptamers harbours sequences that complement to the full sequences of the RNA target. According to Ref. [45], binding of aptamer to their target occasionally happened via three dimensional configurations and involve diverse physiochemical interactions, similar to antibodies. These interactions include electrostatic, hydrophobic, hydrogen-bonding, van der Waals forces, base stacking, and shape complementarity. Moreover, *in vitro* selection of oligonucleotide aptamers against RNA target occasionally resulted in different from the antisense sequence [46]. Hence, we further suggested that our aptamers bound to the RNA target via DNA-RNA stem loop kissing complex.

The hairpin structure of aptamers and RNA target was then generated through mfold algorithm in OligoAnalyzer™ Tool with default parameters from IDT (Integrated DNA Technologies, USA) (Fig. 8B). From several generated hairpin structures, we selected the most appropriate conformation that can form the stem loop kissing complexes by considering for highest number of possible base-pairing. Hairpin structure of PriAseq14 showed a single stem loop, while SeqAseq20 had two loops. The RNA target showed two stem loops at bases position 28th – 32nd and 43rd – 54th. We proposed that PriAseq14 bound to RNA via its stem loop at position 14th to 21st ('ATGATCAC') against the second stem loop of RNA at bases position 47th to 54th ('GUGAUCAU'). The secondary aptamer, SecAseq20 was proposed to bind through its first stem loop at bases position 6th to 10th ('ACCAT') against RNA's first stem loop at bases position 28th to 32nd ('AUGGU').

Development of DNA or RNA aptamers as ligand against RNA target has been reported in several studies particularly in clinical applications such as antiviral therapy [47–49] or detection of viruses like HIV-1 [50] and SARS-CoV-2 [51]. However, detection of target inside plants or originated from plant is still in its infancy even though plants are the

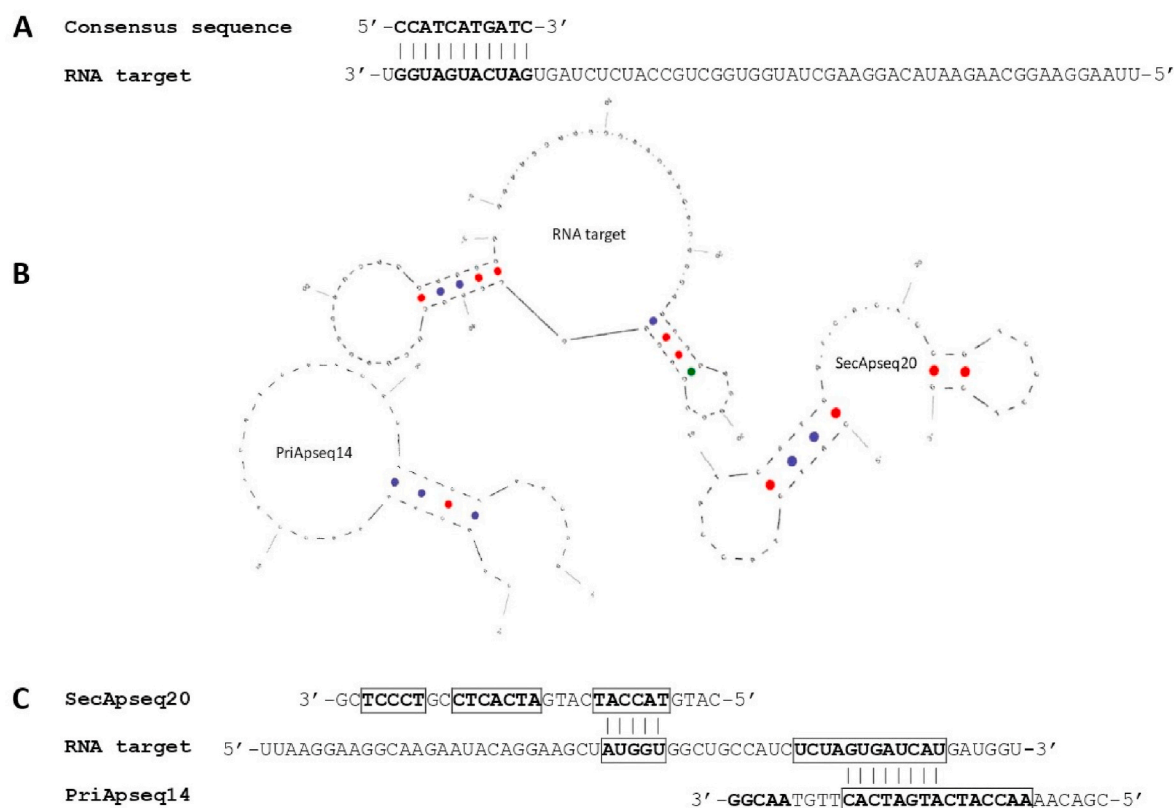


Fig. 8. Proposed complementary base pairing between (A) consensus sequence of aptamers and *EgPIN5* RNA sequence; (C) RNA-DNA loop 'kissing complex' proposed by sandwich-type PriApteq14-RNA-SecApteq20 combination in the lateral flow assay. The base-pairing was proposed from the unpaired bases at the loop of the aptamer-DNA and target-RNA hairpin structures (B). The hairpin structures were generated using OligoAnalyzer™ Tool with default parameters from IDT (Integrated DNA Technologies, USA).

main source of oxygen, food or valued products such as pharmaceuticals, chemicals, or energy fuels [52]. Targets for aptamers in plant biology reported so far are small molecules including phytohormones such as cytokinin [53,54], abscisic acid [55], and salicylic acid [56]. To the best of our knowledge, there is no reported studies regarding application of aptamers for detection of mRNA in plants resulted from response to stress factors. The high specificity and sensitivity features shown by aptamers should be favoured for point-of-care diagnostics in plants instead of laborious PCR method.

5. Conclusions

In this study, we tested *EgPIN5* as mRNA marker in oil palm plantation to support its potential as the plant's biomarker during early infection of *G. boninense*. Through mPCR, the gene was shown to be expressed in symptomless oil palm tree which was believed to be at early stage of *G. boninense* infection. We also developed ssDNA aptamer against the *EgPIN5* RNA fragment in LFA strip and determined the LOD at 2.45 ng/μL of the target. The signal of the LFA strip was visible by naked eye from 12.5 ng/μL to 200 ng/μL. However, low load of RNA copies in field samples resulted in negative results on the LFA strips. We further recommended application of SENSR and isothermal amplification technique which feasible to be carried out on field such as transcription-mediated amplification (TMA) in the future to amplify the low load of the target RNA copies required to meet the amount for LOD and visible signals on the LFA strips. Upon integration with SENSR/isothermal amplification technique, we further proposed our detection technique as surveillance of *G. boninense* infection on field for every two weeks or once per month so that early mitigation of BSR progress can be conducted. Early detection increases the chance of survival rate of the tree and maintain the productivity of the tree. In addition, further

evaluation regarding function of the gene during early response of oil palm towards *G. boninense* infection is highly encouraged. It is hoped that the result of this study can pave the way for utilization of aptamers-based biosensors in point-of-care detection of markers in plant biology.

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Availability of data and materials

The corresponding author will provide the datasets used and analysed during the current work upon reasonable request.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

CRediT authorship contribution statement

Mohammad Nazri Abdul Bahari: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Siti Nor Akmar Abdullah:** Conceptualization, Funding acquisition, Investigation, Methodology, Project

administration, Supervision, Validation, Writing – review & editing. **Nurshafika Mohd Sakeh:** Data curation, Formal analysis, Methodology, Validation, Writing – original draft. **Khairulmazmi Ahmad:** Conceptualization, Supervision. **Noor Azmi Shaharuddin:** Conceptualization, Supervision. **Idris Abu Seman:** Conceptualization, Resources, Supervision. **Rosiah Osman:** Resources, Supervision.

Declaration of competing interest

There are no financial or personal relationships that may be perceived as influencing the authors' work to be declared.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.pmpp.2023.102209>.

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